# MIT Data Analyst 1 Technical Test

2025-03-24

## Introduction

This is a technical test packet for the **Data Analyst 1** position you applied for at **MIT (Massachusetts Institute of Technology)**. Please submit your solutions within 24 hours. We expect this task will take approximately 2 hours of your time, including submission prep. We do not want you to spend more of your time than necessary to complete the test.

The purpose of the test is to establish a minimum technical bar on core skills required for this position. As such, we are evaluating successful completion of the tasks and the professionalism of the code and output. This test data is synthetic and for evaluation use only: no work you complete here will be used by our team for any purpose other than evaluating your application.

You may use any language, IDE, or helper to complete the tasks, but the work you submit should ultimately be your own – you should feel comfortable explaining any part of the code you submit. Additionally, we want to get a feel for not only your answers to the questions, but the manner in which you approach them: to the extent you can document your approaches (including considered but discarded approaches) using in-code documentation, lab notebooks (e.g. Jupyter notebooks or Rmd files), or recordings, that would be useful for us.

You will submit your answers in a GitHub repository, **per the requirements described in Task 4**, which will contain both your code, and a document outlining your results.

The test consists of four tasks:

1. Writing SQL queries (30-45 minutes);
2. Data Analysis and Visualization (30-45 minutes);
3. Sharing a code sample of your previous work (10-15 minutes);
4. Creating a GitHub repository with your code and answers (10-15 minutes).

After sharing your GitHub repository, please respond to us via **email** to confirm submission.

## Task 1: SQL Queries

**Technical Background**:

For this question, you have been given read-only access to a PostgreSQL server, with the following connection details:

- Hostname: `lobbyview.mit.edu`
- Username: `user_analyst`
- Password: `lobbyviewtest`
- Database: `test`

**Please connect to the server as soon as possible to verify that everything works for you**. If you are unable to connect to the server or if you experience networking issues that prevent you from completing the task, reach out immediately. Factors out of your control will not be held against you.

The server contains a single schema called `analyst`, which contains a series of tables, which contain a subset of U.S. lobbying from 2009-2019 (we describe political background below). The tables include data on `bills` (legislation), lobbying `filings` (reports), `registrants`, and `clients`. The schema are designed to be intuitive and self-explanatory, not to trick you. Each table has clearly named key columns to facilitate connecting tables. You may benefit from knowing that the `amount` column in the `filings` table is PostgreSQL's "money" type, which is a non-standard data type that has some properties that may require special handling or conversion.

**Political Background**:

Lobbying is the process where firms and special interest groups interact with elected officials and bureaucrats to share information, learn about legislation that may affect their interests, and advocate for policy outcomes. People who do the work of lobbying are called lobbyists. Lobbyists work for a company called a **registrant**. The company that hires the registrant is called the **client**. Registrants must periodically file activity reports on behalf of clients. These filings include: (a) how much money the client paid the registrant to lobby; (b) which overall issues (tax, immigration, healthcare, etc.) they lobbied on; (c) within those issues, which bills (legislation) they lobbied on. A report has zero or more issues, and an issue has zero or more bills lobbied. Money spent is reported at the filing level, not the issue or bill level.

**Task**:

Please perform all analysis for this task in SQL queries (without using Pandas, R data frames, or other data manipulation tools to perform the task). You may wrap the queries in a script if doing so helps you export your results in a structured way, or submit the queries as `.sql` file in your repository.

(a) Write a query that retrieves registrants whose total lobbying spend exceeds $10,000,000 (we call these "super-registrants"); your query should return all relevant cases, but your document output can be limited to the top 10.

(b) Select the five biggest registrants by dollar amount, and for each registrant, select their five biggest clients (each) by dollar amount in descending order; if any registrant has fewer than five, you may just list their clients.

(c) Among all registrants, identify the registrant who has lobbied the most separate bills in the Medicare/Medicaid issue category. You can identify Medicare/Medicaid lobbying through the issue code MMM. Write a query that returns the unique bill IDs that the registrant has lobbied; your query should return all bill IDs, but your document output can be limited to a sample of 10.

(d) Most bills in the U.S. Congress have a title format that ends in "Act", "Law", or "Resolution". Call these bills "standard titles". Write a query that counts how many bills in the 'bills' table have "standard" titles and how many bills have "non-standard titles". Provide only the totals for each.

The purpose of this task is to evaluate your skills using a variety of core SQL functions, including querying, joining, and string manipulation.

## Task 2: Data Analysis and Visualization

**Technical Background**:

For this question, you will use the same data, but perform analysis on your local computer. You may export the data from the server, or use pre-exported files found in the following Dropbox folder. Do any joining, cleaning, or analysis using local data structures, not SQL.

This question requires visualization(s). We recognize that given time constraints, these will not be publication-ready, but please aim for visualizations that represent the kind of draft you'd present at a weekly stakeholder meeting. Make appropriate decisions about labelling, coloring, spacing, and presentation to yield a clear and attractive figure.

**Political Background**:

In this question, you will consider the role of committees in the U.S. legislative process. In order for a bill (piece of legislation) to become a law, it must first be voted on by a subset of the legislature called a committee. A bill is first referred to a committee; it is debated by the committee, and if the committee approves it, reported out of committee. It is then voted on by the entire legislature. If the bill passes both the U.S. House of Representatives and the U.S. Senate and is signed by the President, it becomes a law. The House and Senate each have separate committees. Committees are organized according to the subject matter they consider. For example, the House of Representatives has a Homeland Security committee and an Energy and Commerce committee. Committees in the Senate often have different names and roles to their House of Representatives counterparts; you may wish to treat the two chambers separately in your analysis below. You can see a full list of committees here. In real life, committees also refer business to sub-committees, but these will be ignored for the test.

**Task**:

Your principal stakeholder has asked you to determine which Committees in the U.S. Congress are the "most important", but has not defined or specified what defines "importance". For the purpose of this task, ignore any external political science literature, past research, or external data – though, we appreciate the enthusiasm! Your job is to answer this open-ended question using the data provided. Your response should target a non-technical reader. You should consider addressing as many of the following goals as possible:

- Propose a definition of "importance" that can be measured based on the data available to you; you have access to the legislation considered by the committees, and lobbying activity around that legislation (in both numbers of reports and dollar amounts).
- Describe the most important committees according to this definition
- Discuss alternative definitions and whether they would change the results
- Evaluate whether the same committees are important across time. You need not use formal statistical tests – a visual or descriptive evaluation is enough.
- Produce at least one visualization of your results; more if necessary to support your analysis.

After your analysis, write one paragraph that describes an additional source of data that you think would help extent this analysis. You do not need to find the actual data (or even verify that it exists at all): simply describe how the proposed data would speak to committee importance.

The purpose of this task is to directly simulate a simplified version of real-world data analysis, including technical writing.

## Task 3: Code Sample

Submit a single script (100+ LOC is a good ballpark to aim for) that you wrote prior to applying for this position. You may choose a script written in any language, and it can be drawn from personal, academic, or professional contexts. The script may, but need not, focus on data analysis. If your portfolio is limited to professional work and non-disclosure agreements prevent you from sharing code, please contact us immediately so we can reach an accommodation.

We are evaluating the submission based on code quality: clarity, documentation or tests if appropriate, adherence to best practices and idioms of your chosen language. We are not evaluating the complexity of the underlying project, so we prefer professional code to clever hacks.

- If you have a public GitHub repository containing a project suitable for this task, you may provide a link to the repository in lieu of submitting the script directly. Please do not link multiple projects or use this as an opportunity to repeat your resumé or portfolio.
- Do not write new code for this task. If you feel it is appropriate you can make readability edits or tweaks to the code, but we want to respect your time.
- If the script is written in collaboration with others, please use documentation (in-code or for Task 4) to identify your own contributions.
- The script does not need to run or compile independently in your repository – you can extract a single component from a larger project you do not share. If the script's role is not immediately clear, please describe its role in the broader project. If the script is a data analysis script, you do not need to share the data.

The purpose of this task is to help you highlight your best work.

## Task 4: Submission

To submit your test, create a GitHub repository and share access with @aaronrudkin and @ryandelano. Upload your code or notebooks in clearly labeled files. Use `.gitignore` if necessary to exclude extraneous project files, environment data such as your login credentials, and any CSV or static data files you used.

The repository should use a `README.md` file to present your results. First, include a brief header identifying the submission packet and, as you see fit, provides an inventory of relevant code files. Below this section, include your results for Tasks 1 and 2 (including tables and visualizations as necessary, and any text answers). Please do a final visual check to ensure that images or formatting is not mangled when viewed on the GitHub web interface.

We will not look through the commit history, branches, etc; do not feel obligated to spend time either cleaning up the repo's history or padding it. Only the final product will be evaluated.

The purpose of this task is to evaluate technical communication, brief technical writing, and use of GitHub.

**Good luck!**